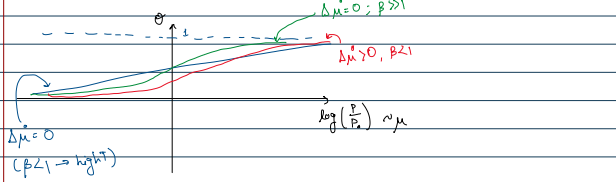
(1) Langmuir : $\frac{\langle N\rangle}{N_s} = \theta = \frac{e^{-\beta(\varepsilon-\mu)}}{1+e^{-\beta(\varepsilon-\mu)}}$ * Ignores interactions



(2) Interactions b/w adsorbates :

→ Use Ising Isotherm : assume interactions have a constant value and only exist b/w nearest neighbors.

→ Can analyze analytically in 1-D : $E = \underbrace{\sum_{i=1}^{N_s} \sigma_i \sigma_{i+1} J}_{\text{Interaction}} + \underbrace{\varepsilon \sum_{i=1}^{N_s} \sigma_i}_{\text{Langmuir part}}$; $\sigma_i = 0, 1$ (Occupied or unoccupied)

J = interaction ($J<0$ → attraction) (is a constant)

As before, $\Xi(\mu, T, N_s) = \sum_{\text{J states}} e^{-\beta(E_j - \mu N_j)}$

Total adsorbates on surface → $= \sum_{N=0}^{N_s} \sum_{\substack{\text{states} \\ \text{w/ } N}} e^{-\beta E + \beta\mu N}$

$= \sum_{N=0}^{N_s} \sum e^{-\beta\left[\sum_{i=1}^{N_s} (\sigma_i \sigma_{i+1} J + \varepsilon\sigma_i - \mu\sigma_i)\right]}$; $\Sigma_{\sigma_i = N}$

Writing as a nested sum:

$= \sum_{\sigma_1=0}^{1} \sum_{\sigma_2=0}^{1} \cdots \sum_{\sigma_{N_s}=0}^{1} e^{-\beta\left[\sum_{i=1}^{N_s} (\sigma_i \sigma_{i+1} J + \frac{\sigma_i + \sigma_{i+1}}{2}(\varepsilon-\mu))\right]}$

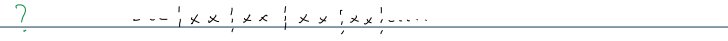
Introduce : $P = \begin{bmatrix} P_{00}: 1 & P_{01}: e^{-\frac{\beta}{2}(\varepsilon-\mu)} \\ P_{10}: e^{-\frac{\beta}{2}(\varepsilon-\mu)} & P_{11}: e^{-\beta(J+\varepsilon-\mu)} \end{bmatrix}$ Symmetric matrices are diagonalizable

(Transfer matrix for lattice statistical mechanics problem)

$\Xi = \sum_{\sigma_1=0}^{1} \sum_{\sigma_2=0}^{1} \cdots \sum_{N_s=0}^{1} \langle\sigma_1|P|\sigma_2\rangle\langle\sigma_2|P|\sigma_3\rangle \cdots \langle\sigma_{N_s}|P|\sigma_1\rangle$; where $\sigma_i = \begin{pmatrix}1\\0\end{pmatrix}$ if "i" is unoccupied

↑ periodic BC's $\sigma_i = \begin{pmatrix}0\\1\end{pmatrix}$ if "i" is occupied

?



Eg : $N_s = 2$,

$\Xi = P_{00}P_{00} + P_{01}P_{10} + P_{10}P_{01} + P_{11}P_{11}$ ($P_{01} = P_{10}$)

sum of the eigenvalues of matrix $P:P = P^2$ ← dyadic product?

P_{00} → element (0,0) of matrix P

In general, Ξ = largest eigenvalue of P^{N_s}

$P^{N_s} = P:P:P:\ldots$ N_s times

$= (\text{eigenvalue of } P)^{N_s}$

$\begin{vmatrix} 1-\lambda & e^{-\frac{\beta}{2}(\varepsilon-\mu)} \\ e^{-\frac{\beta}{2}(\varepsilon-\mu)} & e^{-\beta(J+\varepsilon-\mu)} - \lambda \end{vmatrix} = 0$; $\Xi \approx \lambda_+^{N_s} = \left[\frac{1 + e^{-\beta(J+\varepsilon-\mu)} + \sqrt{(1+e^{-\beta(J+\varepsilon-\mu)})^2 - 4(e^{-\beta(J+\varepsilon-\mu)} - e^{-\beta(\varepsilon-\mu)})}}{2}\right]^{N_s}$

↑ largest evalue

$\langle N\rangle = -\frac{\partial\Omega}{\partial\mu} = \frac{\partial}{\partial\mu}(k_B T \ln \Xi)$

$\therefore \frac{\langle N\rangle}{N_s} = \theta = \frac{e^{-\beta(J+\varepsilon-\mu)} + \frac{-e^{-\beta(J+\varepsilon-\mu)} + e^{-2\beta(\varepsilon+J-\mu)} + 2e^{-\beta(\varepsilon-\mu)}}{\left[(1-e^{-\beta(J+\varepsilon-\mu)})^2 + 4e^{-\beta(\varepsilon-\mu)}\right]^{1/2}}}{1 + e^{-\beta(J+\varepsilon-\mu)} + \left[(1-e^{-\beta(J+\varepsilon-\mu)})^2 + 4e^{-\beta(\varepsilon-\mu)}\right]^{1/2}}$

If $J=0$, $\theta = \frac{e^{-\beta(\varepsilon-\mu)}}{1+e^{-\beta(\varepsilon-\mu)}}$

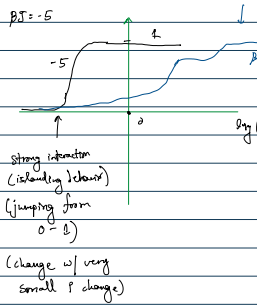
If $|\beta J| << 1$,

$$\theta \simeq \frac{e^{-\beta(\varepsilon-\mu)}}{1+e^{-\beta(\varepsilon-\mu)}}\left[1-\frac{2e^{-\beta(\varepsilon-\mu)}}{(1+e^{-\beta(\varepsilon-\mu)})^2}\beta J\right]$$

→ In 2D, partial analytical solution exists for $\theta = \frac{1}{2}$

Onsager (1942)

→ For more complex interactions, use: + cluster expansion method (power series) + machine learning potentials } Term project idea → different interactions



strong repulsion (Two Langmuir isotherms → Bimodal Langmuir behavior)

S → shape (phase separation)

βJ → order of kT

@ low Pressure → empty / vacant sites
@ P ↑ → filling sites randomly (w/ no nearest neighbor)
@ P = 0.5 → give max no. of adsorbates w/ no nearest neighbors (get a plateau)

Alternate filling



↓

Have to increase the P by a tremendous amount to get more populated sites.

↓

Second Langmuir isotherm (i.e. now you basically have two diff sites)

For 2D case → The binding starts to minimize repulsion. (i.e. the θ will increase by $e^{\beta J}$)



The βJ will increase by 4

↙

given any surface you can intuitively get the shape / plateaus for adsorption

H_2O adsorption on electrodes?
↳ quite attractive H-bonding
↳ complex structure

Def. of monolayer can be different to that of how you define what the surface sites are (and how many bonds they make)